Lab 10

[valid 2024-2025]

Networking
Create an implementation of the Hex game, that allows remote players to start or join games. The application will contain two parts (create a project for each one):

- The **server** is responsible with the game management and mediating the players.
- The **client** will communicate with the server, sending it *commands* such as:
  - create or join a game,
  - submit a move, etc.

The main specifications of the application are:

Compulsory (1p)

- **Create the project ServerApplication**. This will contain (at least) the classes: *GameServer* and *ClientThread*.
- Create the class *GameServer*. An instance of this class will create a *ServerSocket* running at a specified port. The server will receive requests (commands) from clients and it will execute them.
- Create the class *ClientThread*. An instance of this class will be responsible with communicating with a client *Socket*. If the server receives the command *stop* it will stop and will return to the client the response "Server stopped", otherwise it returns: "Server received the request ... ".
- **Create the project ClientApplication**. This will contain (at least) the class: *GameClient*.
- Create the class *GameClient*. An instance of this class will read commands from the keyboard and it will send them to the server. The client stops when it reads from the keyboard the string "exit".

Homework (2p)

- Create the OOP model and implement functionalities of the game.
- The clients will send to the server commands such as: *create game*, *join game*, *submit move*, etc.
- The server is responsible with the game management and mediating the players.
- The games will be played under time control (blitz) with each player having the same amount of time at the beginning of the game. If a player's time runs out, the game is lost.

Bonus (2p)

- Implement an AI (artificial intelligence) for the game. A human player could create a game in which the opponent is the AI, instead of another human player.
- The algorithmic level should be able to determine, at any stage of the game, if at least one path may be formed from one side to another.

Resources

- Slides
- Custom Networking
- Remote Method Invocation (RMI)
- Java Networking

Objectives

- Understand terms and concepts related to networking: protocol, IP, port, URL, socket, and datagram.
- Get familiar with the client-server programming model.
- Write programs that communicate with other programs on the network, using TCP or UDP.
- Get acquainted with RMI technology.